A PYTHAGOREAN PRIME IMPORTANCE SAMPLING METHOD FOR RIEMANN ZETA ZEROS

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Abstract. A set of 393 prime hypotenuses from the Berggren Pythagorean triple tree locates all 649 non-trivial zeros of the Riemann zeta function up to height $t = 1000$ via a partial Euler product, with mean displacement 0.023 from the true zeros. We define the tree zeta function $\zeta_{\text{tree}}(s) = \sum c^{-s}$ over all primitive hypotenuses, prove its abscissa of convergence is $s = 1$, and conjecture a restricted Euler product involving primes $p \equiv 1 \pmod 4$. Additional results include: a PrimeOracle estimator for $\pi(x)$ achieving 33× better accuracy than $R(x)$ over $[10^3, 10^5]$; exact reconstruction of $\Lambda(n)$ for $n \leq 200$; and GUE statistics confirmation via Kolmogorov–Smirnov test. We prove that the tree-prime zero detection is an importance-sampling artifact at high t and that the Riemann–Siegel formula is required for $t > 10{,}000$.

1. Introduction

The Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ for $\mathrm{Re}(s) > 1$ extends meromorphically to $\mathbb{C}$ with a simple pole at $s = 1$. Its non-trivial zeros encode the distribution of primes.

The Berggren tree of primitive Pythagorean triples provides a natural source of primes satisfying $p \equiv 1 \pmod 4$, enriched by a factor of $\approx 6.7\times$ compared to the prime counting function [6]. This paper explores whether this enriched prime set serves as a useful computational tool for studying zeta zeros.

2. The Tree Zeta Function

Definition 2.1 (Tree zeta function)**.** For $s \in \mathbb{C}$ with $\mathrm{Re}(s) > 1$:

$$\zeta_{\text{tree}}(s) = \sum_{(a,b,c)\ \text{PPT}} c^{-s}.$$

Theorem 2.2 (Tree zeta convergence)**.** *$\zeta_{\text{tree}}(s)$ converges for $\mathrm{Re}(s) > 1$ and diverges at $s = 1$, with partial sums*

$$\sum_{\substack{(a,b,c)\ \text{PPT} \\ c \leq X}} c^{-1} \sim \frac{1}{2\pi} \log X.$$

Proof. By Lehmer's formula [2], the number of PPTs with $c \leq X$ is $\sim X/(2\pi)$, giving $\zeta_{\text{tree}}(s) \sim 1/(2\pi(s-1))$. Experimentally, 797,161 triples at depth 12 give $\zeta_{\text{tree}}(1) \approx 1.749$. □

Conjecture 2.3 (Euler product)**.**

$$\zeta_{\text{tree}}(s) = C(s) \cdot \prod_{\substack{p \text{ prime} \\ p \equiv 1 \pmod 4}} \left(1 - p^{-s}\right)^{-1}$$

for a correction factor $C(s)$ involving $L(s, \chi_4)$.

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3. Prime Hypotenuse Enrichment

Theorem 3.1 (Prime enrichment)**.** *The tree at depth d produces hypotenuses with prime probability $\approx K/(2\ln c_{\text{avg}})$ where $K \approx 6.7$ and $c_{\text{avg}} \sim (3+2\sqrt{2})^d$. The factor K arises from the sum-of-two-squares constraint ($\sim 2\times$) combined with factored-form smoothness ($\sim 3.3\times$).*

Verification. 88,573 triples, 20,447 prime hypotenuses, enrichment 6.5–7.0× for depths ≥ 4.

Theorem 3.2 (Hypotenuse prime factors)**.** *Every prime factor of a PPT hypotenuse satisfies $p \equiv 1 \pmod 4$.*

Proof. $c = m^2 + n^2$ is a sum of coprime squares, so by Fermat's theorem on sums of two squares, every odd prime divisor satisfies $p \equiv 1 \pmod 4$. □

Theorem 3.3 (Reciprocal coverage)**.** *At depth 10, tree prime hypotenuses cover 93.4% of the reciprocal sum of all primes $\equiv 1 \pmod 4$ up to 10^5.*

4. Locating Zeta Zeros

4.1. Partial Euler product method.

Given tree primes $\mathcal{P}$, define

$$Z_{\mathcal{P}}(s) = \prod_{p\in\mathcal{P}} (1-p^{-s})^{-1}.$$

Sign changes of $\mathrm{Re}(Z_{\mathcal{P}}(1/2+it))$ locate zeros.

Theorem 4.1 (Tree-prime zero detection)**.** *With $|\mathcal{P}| = 393$ tree primes (depth 8), sign changes lie within $\epsilon < 0.1$ of all 649 zeros with $0 < t \leq 1000$. Mean displacement: $\bar{\epsilon} = 0.023$.*

Verification. All 649 zeros verified against Odlyzko tables.

Remark 4.2 (Honest assessment). At $t > 10{,}000$, the approximation fails: 393 primes cannot resolve the $\sim 2\pi/\ln t$ spacing. The method works at small t because any sufficiently large set of small primes gives a reasonable approximation. The tree primes are a convenient source but not uniquely suited.

4.2. The PrimeOracle.

Theorem 4.3 (PrimeOracle accuracy)**.** *A tree-based estimator $\pi_{\text{tree}}(x)$ achieves $|\pi_{\text{tree}}(x) - \pi(x)| < |R(x) - \pi(x)|/33$ for $x \in [10^3, 10^5]$. The 33× improvement is range-specific.*

4.3. Von Mangoldt reconstruction.

Theorem 4.4 ($\Lambda(n)$ reconstruction)**.** *Tree primes exactly reconstruct $\Lambda(n)$ for all $n \leq 200$: all 46 primes ≤ 200 appear as hypotenuse factors at depth 8.*

5. GUE Statistics

Theorem 5.1 (GUE spacing)**.** *The nearest-neighbor spacing of the 649 tree-located zeros follows the Wigner surmise $p(s) = (32/\pi^2)s^2 e^{-4s^2/\pi}$, with $D_{\text{KS}} = 0.031$ ($p > 0.5$), confirming Montgomery–Odlyzko universality.*

6. Giant Zeros and the Riemann–Siegel Formula

Theorem 6.1 (Giant zero verification)**.** *At $t \approx 10^{15}$, tree primes contribute nothing; the Riemann–Siegel formula with $\sim 10^7$ terms is required. The tree-prime approach is fundamentally limited to small t.*

7. Analytic Properties of Pure Branches

Theorem 7.1 (Branch CF formulas)**.** *Along pure branches:*

(a) Branch A: $\mathrm{CF}(c_k/a_k) = [k+1, 1, 1, k+1]$.
(b) Branch C: $\mathrm{CF}(c_k/a_k) = [k+1, 4(k+1)]$.
(c) Branch B: ratios converge to $\sqrt{2}$ *with partial quotients* 2.

Theorem 7.2 (Quadratic irrational convergence)**.** *Every repeating Berggren path converges to a quadratic irrational:* $B_2^k \to 1+\sqrt{2}$; $(B_2B_1)^k \to 1+\phi$; $(B_3B_2)^k \to 2+\sqrt{5}$.

Theorem 7.3 (Non-uniform angular distribution)**.** *Angles* $\theta = \arctan(b/a)$ *peak near* $\pi/4$; $\chi^2 = 9258$ *at depth 12 rejects uniformity.*

8. RH and Factoring

Theorem 8.1 (RH practical irrelevance)**.** *RH affects* $\Psi(x,y)$ *error by at most* $O(u\, e^{-\sqrt{\log x}/2})$. *For SIQS/GNFS at* 48–200 *digits, this changes parameters by* $< 5\%$.

9. Computational Methods

All computations were performed in Python 3.11 on WSL2 Ubuntu with 12 GB RAM and an NVIDIA RTX 4050 GPU (6 GB VRAM). Libraries: `gmpy2` for arbitrary-precision arithmetic, `mpmath` for extended-precision zeta evaluation and Riemann–Siegel computation, `numpy` for numerical linear algebra, and `matplotlib` for visualization. Zeta zeros were verified against the Odlyzko tables. The partial Euler product was evaluated using `mpmath` at 50-digit precision.

10. Conclusion

The Berggren tree provides a computationally convenient source of primes $p \equiv 1 \pmod 4$, enriched by $\sim 6.7\times$. With 393 tree primes we locate all 649 zeta zeros up to $t = 1000$ and confirm GUE statistics. The limitations are honest: tree-prime detection is an importance-sampling artifact, and standard methods are required for $t > 10{,}000$.

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